Since each wave pattern carries the same amount of energy, an infinite number of them translates into an infinite amount of energy. At the turn of the century, there was a gargantuan fly in the theoretical ointment.

Making Lumps at the Turn of the Century

In 1900 Planck made an inspired guess that allowed a way out of this puzzle and would earn him the 1918 Nobel Prize in physics.2 To get a feel for his resolution, imagine that you and a huge crowd of people—"infinite" in number—are crammed into a large, cold warehouse run by a miserly landlord. There is a fancy digital thermostat on the wall that controls the temperature but you are shocked when you discover the charges that the landlord levies for heat. If the thermostat is set to 50 degrees Fahrenheit everyone must give the landlord $50. If it is set to 55 degrees everyone must pay $55, and so on. You realize that since you are sharing the warehouse with an infinite number of companions, the landlord will earn an infinite amount of money if you turn on the heat at all.

But on closer reading of the landlord's rules of payment you see a loophole. Because the landlord is a very busy man he does not want to give change, especially not to an infinite number of individual tenants. So he works on an honor system. Those who can pay exactly what they owe, do so. Otherwise, they pay only as much as they can without requiring change. And so, wanting to involve everyone but wanting to avoid the exorbitant charges for heat, you compel your comrades to organize the wealth of the group in the following manner: One person carries all of the pennies, one person carries all of the nickels, one carries all of the dimes, one carries all of the quarters, and so on through dollar bills, five-dollar bills, ten-dollar bills, twenties, fifties, hundreds, thousands, and ever larger (and unfamiliar) denominations. You brazenly set the thermostat to 80 degrees and await the landlord's arrival. When he does come, the person carrying pennies goes to pay first and turns over 8,000. The person carrying nickels then turns over 1,600 of them, the person carrying dimes turns over 800, the person with quarters turns over 320, the person with dollars gives the landlord 80, the person with five-dollar bills turns over 16, the person with ten-dollar bills gives him 8, the person with twenties gives him 4, and the person with fifties hands over one (since 2 fifty-dollar bills would